Proof and Resolution: A Formal Framework for Human Agency in Machine Mathematics

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Abstract—*Automated proof systems, artificial intelligence pipelines, and formal verifiers have fundamentally changed the scale and speed of mathematical proof production. However, mechanical verification that a formal expression holds within a logical system is fundamentally distinct from satisfying a human mathematical requirement. This paper establishes a comprehensive framework separating Proof, a formal truth-preserving relation, from Resolution, a requirement-relative acceptance relation. We develop the concepts of Situational Reality, Granularity, and the Resolution Boundary to provide a mathematically rigorous vocabulary for evaluating machine contributions. Crucially, we formalise an explicit, eight-step mathematical Resolution Protocol designed to ensure absolute rigor, semantic correspondence, and human comprehension. This framework provides a structural mechanism for the mathematical community to integrate automated results safely without surrendering human agency, insight, or philosophical ownership.*

1. Introduction

Contemporary breakthroughs in machine intelligence, reinforcement learning, and auto-formalisation pipelines—exemplified by systems such as DeepMind's AlphaProof—have brought mathematics to a critical historical inflection point. Logical systems are now capable of independently charting massive proof search spaces, generating complex mathematical proof terms, and mechanically checking their formal correctness via kernel architectures like Lean. This transition eliminates the traditional scaling limit of human verification, opening the door to an unprecedented expansion of verified mathematical data.

Yet, this technological leap exposes a deep epistemological rift. A machine kernel can safely guarantee that a formal statement is derivable from a set of declared axioms, but it remains structurally blind to the wider mathematical context. It cannot verify if the formalised expression accurately represents the original human intuition, whether an accidental over-constraint has rendered the result trivial, or whether the resulting proof term conveys any actual mathematical explanation. As the density of machine-generated mathematics increases, the community faces the danger of accumulating a vast sea of mechanically certified truths that are completely unreadable, uninstructive, and conceptually detached from human insight.

This paper resolves this tension not by opposing mechanical verification, but by precisely delineating its boundaries. We assert that many current debates confuse two entirely distinct epistemic operations: *Proof* and *Resolution*. Proof is a formal, proposition-directed derivation bounded by an explicit logical calculus. Resolution is a requirement-directed, observer-relative reconciliation that establishes whether a given requirement has

been satisfactorily addressed within an explicitly defined **Situational Reality**. By formalising this distinction, we provide a concrete framework where automated proof acts as an exceptionally strong contributor to human knowledge, without masquerading as the entirety of mathematical practice.

2. Situational Reality and Granularity

No mathematical reasoning, software execution, or scientific model operates on reality in its entirety; instead, they operate on a carefully selected subset of distinctions. We define **Situational Reality** as the bounded set of distinctions, observations, propositions, and trusted behaviours that an observer accepts as meaningful for a requirement, at a declared granularity. Letting O denote an observer and R denote a requirement, the Situational Reality $SR(O,R)$ isolates the narrow, operational reality upon which reasoning actually proceeds.

This boundaries-first approach relies heavily on **Granularity (G)**, which is the explicitly selected level at which distinctions are treated as primitive or collapsed. A conceptual projection can be formalised as $\pi_G : W \to SR_G$, where W is the richer uncollapsed world and SR_G is the exposed reality. Under this model, an entity whose internal distinations are deliberately placed below the current granularity is treated as a **Black Box**. Its internal mechanics are collapsed and replaced by an accepted observable contract or **Trust Point**, denoted as $T_G(x)$. This design prevents infinite recursive reopening of foundational assumptions, allowing engineering and mathematics to successfully terminate.

3. The Structural Separation of Proof and Resolution

Proof and requirement satisfaction are frequently treated as interchangeable metrics of certainty. This is a foundational error. A formal proof establishes a rigid logical derivation: $A \blacksquare_G P$, declaring that given an accepted formal system and premises A at granularity G, the proposition P logically follows. Proof moves unidirectionally from accepted premises to a proposition. It is strictly proposition-directed.

In contrast, Resolution begins with a requirement R and evaluates whether the observer's available Situational Reality satisfactorily addresses it. It can be written as the acceptance relation $SR_{O,R}^{G,T} \Rightarrow \checkmark$. Resolution is requirement-directed and structurally omnivorous; it consumes heterogeneous forms of evidence, including formal proofs, empirical measurements, and human validations. While a formal proof can be consumed as an incredibly robust Trust Point within a Situational Reality, the proof itself does not imply Resolution of the motivating requirement. A proof can be fully closed by a machine, while the broader mathematical resolution remains entirely open.

4. The 8-Step Mathematical Resolution Protocol

To operationalise this framework within contemporary research pipelines, we present a robust, mathematically bulletproof Resolution Protocol. This protocol transforms the vague question of 'whether to trust an AI result' into a structured, auditable, and multi-dimensional verification process. It guarantees that formal soundness is harmonised with semantic correspondence and human insight.

Step 1: Requirement Specification (R) — Explicitly declare the observer's core objective. Is the requirement strict mechanical correctness, informal conjecture resolution, pedagogical explanation, or publication-worthy discovery? The requirement dictates the necessary resolution boundary and granularity.

Step 2: Formal Object Identification (F) — Isolate the exact, uncompiled formal syntax string or proof term generated by the machine. This ensures that the object of inquiry is uniquely identified and locked against dynamic alterations.

Step 3: Semantic Correspondence Validation (C) — Audit the mapping between the natural-language problem *Conjecture* and the formal statement *F*. This step verifies that *F* is a faithful translation of *Conjecture*, ensuring the machine did not accidentally solve a trivialized, over-constrained, or distorted variation of the target problem.

Step 4: Boundary & Assumption Enumeration (A) — Completely list every axiom, core definition, imported library, and foundation dependency relied upon by the proof. Any unvetted library or external lemma must be flagged as an open risk rather than an accepted premise.

Step 5: Kernel Verification (V) — Subject the formal object *F* and its proof term to independent verification by a trusted, isolated logical kernel (e.g., Lean, Coq). This produces the foundational truth certificate: $A \blacksquare F$, closing the mechanical correctness requirement.

Step 6: Provenance & Attribution Tracking (P) — Document the complete structural lineage of the proof. This maps the exact contributions of human prompt engineers, language model variations, reinforcement learning loops, and pre-existing library templates, establishing research integrity and intellectual lineage.

Step 7: Explanatory Abstraction & Interpretation (I) — Translate the low-level machine proof term into a higher-level conceptual representation. If the requirement demands human understanding, this step extracts the core mathematical mechanism, structural insight, or novel technique from the machine's automated search.

Step 8: Final Acceptance & Trust Mapping (T) — Reconcile all preceding steps against the original requirement *R*. The observer explicitly maps which components are accepted as static Trust Points and outputs a tri-state resolution value: *Y* (Resolved/Accepted), *N* (Rejected), or *U* (Unresolved/Open).

5. Application and Philosophical Value

By enforcing this 8-step protocol, the mathematical community can systematically process AI-generated breakthroughs. For instance, when an AI system outputs an exceptionally dense proof of a long-standing conjecture, a journal's editorial board can accept the kernel verification (Step 5) as absolute proof of formal correctness, while keeping the overall resolution state as *Unresolved (U)* until Semantic Correspondence (Step 3) and Explanatory Abstraction (Step 7) are completed by human referees. This structural approach eliminates the false binary between blind technophilia and defensive machine rejection, defining a scalable, permanent role for human mathematical judgement.

6. Conclusion

Human and computational reasoning cannot avoid operating within bounded granularities; what can be avoided is leaving those boundaries implicit. Proof preserves truth within an isolated formal system, but Resolution reconciles a requirement with a structured, multi-dimensional reality. The mathematician's primary responsibility in the age of artificial intelligence is not to race automated proof searches in speed or scale, but to establish, audit, and maintain the Resolution Boundary. The 8-step protocol presented herein offers a rigid, comprehensive mechanism to safeguard mathematical rigor while ensuring that the discipline remains a deeply meaningful, human-centered endeavor.